GenTwin: Using Generative AI and Digital Twins to Uncover Hidden Cybersecurity Gaps in Industrial Systems

Background

The Secure Water Treatment (SWaT) testbed represents a real-world industrial control system used to operate and monitor a water treatment plant. It captures both normal operational behavior and multiple cyber-attack scenarios.

In critical infrastructure environments, traditional anomaly detection techniques often fail to detect unknown attack vectors or configuration-based vulnerabilities.

Digital Twins (DTs) provide virtual replicas of physical processes, enabling simulation, prediction, and analysis of system behavior. Meanwhile, Generative Artificial Intelligence (GenAI) models can learn complex operational patterns from historical data and generate synthetic attack or anomaly scenarios.

The convergence of these technologies enables a new paradigm: a *cyber-aware digital twin* capable of proactively identifying vulnerabilities and testing attack scenarios before adversaries exploit them.

The Challenge

Participants are tasked with building an innovative prototype or conceptual solution using the SWaT dataset to demonstrate how Generative AI and Digital Twin technologies can be combined to:

1. Identify hidden cybersecurity gaps or vulnerabilities in the water treatment process.
2. Simulate and visualize attack impacts within a digital twin environment.
3. Propose detection or mitigation strategies that enhance overall system resilience.

Expected Directions

Teams may choose one or more of the following directions:

- **GenAI Modeling**: Use Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), or diffusion models to generate previously unseen attack patterns based on SWaT data.
- **Digital Twin Simulation**: Develop a simplified digital twin of the SWaT process using tools such as SimPy, AnyLogic, MATLAB, or custom simulation logic.
- **Gap Discovery**: Correlate generated anomalies with process-level outcomes to identify weak points, missing controls, or unsafe system states.
- **Cyber Defense Dashboard**: Build a visualization or decision-support dashboard highlighting predicted vulnerabilities and recommended mitigation actions.

Deliverables

1. A working prototype or conceptual demonstration showing how GenAI and Digital Twin technologies interact for cybersecurity gap identification.
2. A short presentation or dashboard summarizing:
   - How the model was trained or the system was simulated.
   - Identified vulnerabilities or security gaps.
   - Proposed improvements to detection mechanisms or system resilience.

Evaluation Criteria

Criterion	**Weight**	**Description**
Innovation	30%	Novelty in combining GenAI and Digital Twin concepts
Technical Depth	25%	Soundness of modeling or twin implementation
Cybersecurity Impact	25%	Clarity and significance of identified gaps
Usability & Visualization	10%	Clarity of outputs and insights
Presentation & Storytelling	10%	Effectiveness of communication and narrative

References

[1] iTrust Centre for Research in Cyber Security, Singapore University of Technology and Design, *Secure Water Treatment (SWaT) Dataset*.
Available at: `https://itrust.sutd.edu.sg/itrust-labs_datasets/dataset_info/`

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